

Socheat Chea

AI Builder

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I build production software with AI coding agents, and I run it myself. Claude Code writes the code, Codex audits it independently, and I write the specs, set the guardrails, make the decisions and do every deploy. In 2026 this process put **five systems into production**, from a real-time crypto trading platform to a loyalty program used by a salon’s customers.

Before this I spent eight years in business roles: B2B sales for a corporate e-learning company, market research for clients around the world, digital marketing, and hotel online sales. I hold an MBA in International Business. That is why my AI work starts from a real business problem and ends with something people can use every day.

5 systems live in production	1,000+ commits across 7 product repos	6,500+ automated test functions	200+ independent AI audit verdicts
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HOW I WORK

One AI builds, another AI audits, and I stay in charge

Every change starts as a written plan and passes two independent checks before it reaches real users.

- 1

Write the card. ME
Scope, what must not change, and the bar it has to pass. Written before any code.
- 2

Pre-build audit. CODEX
Reads the real code and the card, then returns PASS PASS WITH CONDITIONS or FAIL.
- 3

Build. CLAUDE CODE
Implements the change switched off by default, adds tests, and hands back a diff with a report.
- 4

Post-build audit. CODEX
Checks the diff and every claim in the report. A FAIL sends it back to step 3.

5 Deploy and watch. [ME](#)

I deploy and switch features on by hand, then check the live data.

The auditor can't edit code.

Codex can only write to a folder outside every repository. It's a sandbox setting, not a request.

The builder can't touch production.

Claude Code is blocked from pushing code, logging into servers, restarting services or reading secret files.

Agreement isn't proof.

Two AIs agreeing is two opinions. Changes to live behaviour are proven with real forward data.

REAL EXAMPLES FROM THE REVIEW RECORD

FAIL Claude's tests mocked out the code under test

So a real crash never ran during testing. Its report also made a false claim. Fixed and passed in round 2.

FAIL A safety limit failed open after a restart

During a database outage the limit stopped working, even though all 69 tests were green. Fixed before release.

FAIL A payment QR code broke on one Python version

Caught because the tests run on two Python versions. Production used the one that broke.

More than half of first-round audits come back FAIL. That is the process working. Used on 3 projects with 130+ planned changes in total.

SELECTED WORK

Six systems, five of them live

Search memory for AI coding agents (RAG)

● **Working, measured** Oct 2026 720 documents · 55 test questions · 458 tests

A private search engine over 720 of my own documents, so Claude Code and Codex find past decisions before they build. Five search methods, measured on 55 questions frozen before the engine's search code.

- **Measured on held-out questions.** On 9 English in-scope questions held out from tuning, the chosen method matched the answer key in its top 5 section groups for 7 of them (77.8%). The target was 8.
- **Five search methods** compared: meaning search, keyword search (BM25), the two merged by rank, a router that uses the merged list when a question names an ID, and a cross-encoder reranker, chosen by a rule written before the first run.
- **Local and private.** PostgreSQL with pgvector on my own PC; no cloud database and no outside embedding API. Before storage, pattern-based rules leave out files that look like credentials or customer records and blank out recognised secrets; they can miss things.

- **Audited at every step.** Codex audited each of the six build cards: 13 rounds, 11 FAIL and 2 PASS. Each FAIL ended in a fix or a decision I recorded.

HARDEST PROBLEM

Tuning on the questions I report on made search look better than it is. Improving search raised the top-5 score on the 29 open questions from 79.3% to 86.2%, with more text in each result. On 9 questions held out from tuning it scored 77.8%: close to the untuned run, but on different questions, and one short of my target. Without the sealed set, I would have reported 86.2%.

Tech: Python, PostgreSQL + pgvector, bge-m3, BM25, bge-reranker-v2-m3, pytest

K+ crypto trading platform

● **Live, real money** Mar-Sep 2026 480 commits · ~111k lines · ~4,000 tests

Three services that stream live order data from four crypto exchanges (Binance, Bybit, OKX, Hyperliquid), score it with a rule engine, and place real trades on Bybit. A demo copy runs the same code, so every change is tested on demo before it goes live.

- **Data collector** for 7 live exchange feeds and 6 coins, updated every 5 seconds and served over a signed private API.
- **Signal engine** that checks every trade idea against a chain of rules. Restructured from one 3,858-line file into six packages, with parity tests checking that its decisions stayed identical.
- **Trade executor** with safety first: every entry is followed straight away by its stop-loss, with a market close as the planned response if the stop is rejected, plus a kill switch and checks against the exchange every 30 seconds.
- **A/B test harness** that pairs demo and live trades and compares them with a proper statistical test.

HARDEST PROBLEM

The data collectors were silently dropping trades. Five of seven feeds kept only the last trade in each batch from the exchange. I had counters added first to measure it: **65–81% of trading volume** on two exchanges was being lost. After the fix, all older data was marked as pre-fix and kept out of any recalibration.

Tech: Python (asyncio), WebSockets, FastAPI, SQLite, pandas, Bybit API, PM2, cron, Telegram alerts

LinkedIn content agent

● **Live, weekly** Sep 2026, 10 days to launch 44 commits · 100 tests

An AI agent that drafts LinkedIn posts teaching Cambodian business leaders how to use AI in real work. It writes each post and its image, checks them, and sends them to me in Telegram. Nothing is published until I approve it. My LinkedIn posts about AI at work are drafted by this agent.

- **Structured output.** The model must return a strict JSON format, and every answer is validated before use.
- **Automatic quality checks.** Rules catch made-up stories, promised results and sales language. A failed draft goes back to the model with the exact problems, up to three tries.
- **Facts with sources.** Every number in a post is matched against a list of sourced facts and shown to me before I approve.

- **Contained AI.** The model runs locked down, with no internet tools and no access to publishing passwords.

Tech: Python, Pydantic, Codex CLI (first version on the Claude API), Telegram Bot API, SQLite, Typefully

Salon loyalty bot

ខ្មែរ · EN

- **Live with customers** Aug–Sep 2026, live within six weeks 303 commits · 4,124 tests

A Telegram loyalty program that replaced paper stamp cards for a salon in Cambodia. Staff give points at the counter. Customers check their points, rewards and a personal QR code in Khmer or English.

- **A points record that can't be edited.** The database itself blocks changes and deletions, and corrections are added as new entries, like accounting.
- **A simple staff flow:** enter the amount, preview, confirm.
- **Khmer first.** All Khmer text is written and approved by the owner.
- **Tested hard.** Deliberate bugs are planted after each change to prove the tests catch them. The last two changes caught 77 of 77 and 34 of 34.

Tech: Python, aiogram, PostgreSQL, SQLAlchemy, Alembic, pytest, GitHub Actions

Ball Khmer football bot

ខ្មែរ · EN

- **Live** Feb–Aug 2026 201 commits · 845 tests

A Telegram bot for Cambodian football fans with live scores, fixtures, a prediction game, match alerts and a paid VIP tier, in Khmer and English.

- **Prediction game** with points, streaks, badges, a monthly leaderboard and referral codes.
- **Live alerts** checked every 60 seconds during matches and sent when a score changes, and a morning preview in each user's language.
- **Prediction tracking.** Match predictions come from an outside ML service. The bot saves them before kickoff and records results after, so accuracy can be measured honestly over time.

HARDEST PROBLEM

When the data provider failed, the bot kept the empty answer, so a busy day could show “No matches today” and the 8 AM preview could reach every user without its predictions. Now it says the data is unavailable instead of guessing, and the audit caught two more timing bugs before release.

Tech: Python, python-telegram-bot, SQLAlchemy, SQLite, Redis, APScheduler

Crypto news brief

● **Live** Sep–Oct 2026 37 commits · 283 tests

A private Telegram news service: two daily digests plus 24/7 alerts on hacks, exchange outages and US regulation, collected from 11 free sources with a source attached to every item.

- **No AI where it isn’t needed.** Simple rules handle selection today. An AI model will be added only when a test shows it improves the result enough to justify the cost.
- **Tested on real output.** When a review of live digests failed on quality, version 3 fixed it by merging duplicate reports of the same incident into one story.

Tech: Python, SQLite, systemd timers, Telegram

HOW I TEST IDEAS

Plan the test first, keep the “no” answers

Before changing a live system I write down the test in advance: what I expect, how I’ll measure it, and what result would prove me wrong. Most answers were “no”, and keeping them stopped bad changes from going live.

QUESTION	METHOD	RESULT
Can smarter profit targets fix exits?	121 rules tested on separate past and future data	0 of 121 passed
Is stop-loss placement the problem?	Studied every winning and losing trade, replayed 7 rules	Stops were fine
Does the one rule that survived hold up live?	Ran it on demo with an automatic kill switch, then A/B against live	Retired
Do entries come too late after a market shift?	Statistical timing test on 162 signals	No effect

EXPERIENCE

Business roles before building with AI

Jan 2022 – now	Founder The Knowledge Plus · Cambodia Crypto market education: daily videos, chart explanations and training. Since 2026, I’ve also built and run the K+ trading platform and the bots above.
Jan 2023 – now	Digital Specialist Ti-Phsar · Cambodia Websites, social media, paid ads and hotel yield management for clients.

Jan 2020 – Apr 2025

Freelance Market Researcher

Ravenry · Cambodia

Competitor research, data gathering, marketing analysis and content writing for clients around the world.

Nov 2019 – Nov 2020

Business Development Representative

Speexx · Thailand

B2B sales for a corporate e-learning company. Found decision-makers, set up meetings with company leaders, and handled negotiation using HubSpot, SalesLoft and LinkedIn Sales Navigator.

Jan 2018 – Sep 2019

Market Research Analyst (part-time and internship)

Canvassco · Thailand

Desk research, competitor and trend reports in English, and data analysis in Excel and PowerPoint.

Jun 2017 – Nov 2018

Social Media and Online Marketing Assistant Manager

Chan Angkor Boutique · Siem Reap

Online and seasonal promotions, and hotel booking channels (Expedia, Booking.com, Agoda, Airbnb).

EDUCATION

Degrees and training

MBA, International Business

Panyapiwat Institute of Management, Thailand ·
2020–2022

BBA, International Business

Dhurakij Pundit University, Thailand · 2016–2019

Information Technology training

University of South-East Asia · 2015–2016

English for General Education

Australian Centre for Education · 2014–2016

SKILLS

What I work with

AI coding agents

Claude Code, Codex CLI, project instructions (CLAUDE.md, AGENTS.md), custom skills, sandbox and permission setup, builder–auditor review loops

LLM applications

Structured JSON output, automatic validation and retry, human approval steps, prompt caching, running models safely

Backend

Python, FastAPI, Telegram bots (aiogram, python-telegram-bot), PostgreSQL, SQLite, Redis, SQLAlchemy, REST and WebSocket APIs

Operations

Linux servers, PM2, systemd, cron, GitHub Actions, backups, monitoring and alerts

Testing and data

pytest, mutation testing, A/B testing, statistics, pandas, Excel, market research

Business

B2B sales and negotiation, digital marketing, customer relationships, staff training

LANGUAGES

I work in English

English

Full professional

Khmer (ខ្មែរ)

Native

Thai

Elementary

ABOUT THE CODE

My code repositories are private because they hold live trading systems and client data. I'm happy to walk you through any project on a screen share, including the plans, audit results, tests and deploy history. You can also try the Ball Khmer bot live in Telegram at @Ball_khmerbot.